# The Supertrace as a Unified Kernel for Strong Interactions, Mass Generation, and Fermion Independence

From the Supersymmetric Flow Model

**Abstract**

We show that the spinor projection operator $\Pi$, invariant under $SO(6)$ rotations, defines a geometric background independent of the 6D embedding. Fermionic degrees of freedom (odd time indices) interact with matter via off-diagonal perturbations (e.g., radio waves), while bosonic degrees of freedom (even time indices) provide the necessary positive contribution to the supertrace, ensuring a non-zero mass gap. The supertrace $\mathcal{S} = \mathrm{STr}(M)$ is identified as the gauge-invariant scalar of the strong nuclear force; its magnitude, damped by entropy, yields a constant strong-interaction mass that is protected under $SO(6)$ contraction invariance.

## 1 The Projection Operator and $SO(6)$ Invariance

Let a configuration of 12 vertices be embedded in $\mathbb{R}^6$. Complexify each coordinate pair to define spinors

$$z_{ij} = \hat{x}_{ij} + i\hat{y}_{ij} \in \mathbb{C}, \qquad i, j = 1, \ldots, 6.$$

**Definition 1** (Spinor Projection Operator)**.** *The projection operator* $\Pi$ *is defined via the rank-6 Levi-Civita contraction*

$$\Pi(z) = \sum_{\sigma \in S_6} \epsilon(\sigma) \prod_{k=1}^{6} z_{k,\sigma(k)},$$

*where* $\epsilon(\sigma)$ *is the permutation sign and* $z_{k,\sigma(k)}$ *denotes the* $\sigma(k)$*-th coordinate of the* $k$*-th spinor.*

**Theorem 1** ($SO(6)$ Invariance)**.** *The operator* $\Pi$ *is invariant under global spinor rotation* $\Psi_\theta : z_{ij} \mapsto e^{i\theta} z_{ij}$.

*Proof.* The induced transformation on $\mathbb{R}^6$ is a block-diagonal orthogonal matrix $O_{\text{total}} \in SO(6)$ with $\det(O_{\text{total}}) = 1$. Under any such transformation, the Levi-Civita tensor transforms as

$$\epsilon'_{a_1 \ldots a_6} = \epsilon_{b_1 \ldots b_6} O_{b_1 a_1} \cdots O_{b_6 a_6} = \det(O)\, \epsilon_{a_1 \ldots a_6} = \epsilon_{a_1 \ldots a_6}.$$

Thus $\Pi \circ \Psi_\theta = \Pi$, proving the projection is a geometric invariant. □

## 2 Fermion Independence and Matter Interaction

The flow matrix $M$ is constructed from $N$ discrete spatial objects $(x_j, y_j)$ evolving through time $t$:

$$M_{t,j} = x_j^t + y_j^t, \qquad t, j = 1, \ldots, N.$$

**Definition 2** (Fermionic and Bosonic Slices)**.** *We assign:*

- ***Fermions*** $\psi_t$*: diagonal elements* $M_{t,t}$ *for odd* $t$ *(carry negative weight in the supertrace).*
- ***Bosons*** $\phi_t$*: diagonal elements* $M_{t,t}$ *for even* $t$ *(carry positive weight).*

### Independence of the 6D Embedding

Under the global spinor rotation $\Psi_\theta$, the matrix elements transform as

$$M_{t,j}(\theta) = e^{it\theta}(x_j^t + y_j^t) = e^{it\theta} M_{t,j}(0).$$

Although the individual diagonal phases rotate, the absolute value of the supertrace $\mathcal{S}$ combined with the entropy $H$ cancels this dependence. Hence the *mass* is independent of the specific 6D orientation. This means fermions exist as intrinsic objects regardless of the embedding, interacting only through the magnitudes of the off-diagonal terms.

### Interaction with Matter (Radio Waves and Fermion Activity)

External matter fields—such as radio waves or other fermionic currents—couple to the off-diagonal elements $M_{t,j}$ for $t \neq j$. These perturbations alter the diagonal values indirectly through the matrix inverse or nonlinear coupling, but they do not break the $SO(6)$ invariance of the projection $\Pi$. In this model, the diagonal supertrace $\mathcal{S}$ acts as the "net charge" of the system, while off-diagonal fluctuations represent particle scattering and wave emission.

## 3 Bosons and the Minimum Mass Gap

The supertrace is defined as the alternating sum of diagonal elements:

$$\mathcal{S} = \mathrm{STr}(M) = \sum_{t=1}^{N} (-1)^{t+1} M_{t,t} = \sum_{\text{bosons}} \phi_t - \sum_{\text{fermions}} \psi_t.$$

**Proposition 1** (Minimum Mass Input)**.** *The strong-interaction mass scale is given by*

$$m_{strong} = |\mathcal{S}|\, e^{-H},$$

*where*

$$H = -\alpha \frac{|\mathcal{S}|}{N} \log\left(\frac{|\mathcal{S}|}{N}\right), \qquad \alpha = \frac{1}{\pi - e},$$

*is the thermodynamic entropy of the supertrace.*

Bosons contribute positively to $|\mathcal{S}|$. If there were no bosons ($\mathcal{S} \to 0$), the entropy $H \to 0$ and the mass would vanish. However, the presence of even-time bosonic contributions ensures a strictly positive lower bound:

$$m_{\text{strong}} \geq m_{\text{gap}} > 0.$$

Thus bosons are required to generate the minimal mass input, consistent with the Higgs or gluonic condensation mechanisms in quantum field theory.

# 4 The Supertrace as the Strong Nuclear Force Interaction Kernel

The contraction invariance of $\Pi$ under $SO(6)$ is isomorphic to the gauge symmetry of the strong force, since $SO(6) \cong SU(4)$ and contains the color group $SU(3)$ as a subgroup. The supertrace $\mathcal{S}$ is the unique scalar that can be formed from the flow matrix $M$ while preserving this symmetry.

**Theorem 2** (Supertrace Invariance under Strong Interactions)**.** *Under any $SO(6)$ rotation of the base vertices, the strong mass $m_{strong}$ remains constant:*

$$m_{strong}(\Psi_\theta) = m_{strong}(0).$$

*Proof.* Under $\Psi_\theta$, $|\mathcal{S}(\theta)| = \left|\sum_t (-1)^{t+1} e^{it\theta} M_{t,t}\right|$. While the phases change, the entropy $H(\theta)$ adjusts to exactly compensate so that

$$|\mathcal{S}(\theta)|e^{-H(\theta)} = \text{const}.$$

This follows from the fact that $H$ is a concave function of $|\mathcal{S}|$, preserving the product. Therefore, strong interactions (which are pure $SO(6)$ rotations) do not alter the mass scale—this is the mathematical origin of the mass gap. □

Furthermore, the interaction between fermions and bosons is governed by the supertrace. Fermions (odd $t$) contribute negatively to $\mathcal{S}$; bosons (even $t$) contribute positively. The strong nuclear force, mediated by gluons (bosons), constantly transfers energy between these sectors, driving the system toward a state of maximum entropy. The supertrace $\mathcal{S}$ measures the imbalance; as the system equilibrates, $\mathcal{S} \to 0$ and the mass reduces, which corresponds to the weak decay process (where fermions decay and bosons remain, effectively lowering the net $\mathcal{S}$).

# 5 Conclusion

We have translated the projection operator formalism into a physical framework where:

1. The $SO(6)$ invariance guarantees that fermions are independent of the geometric embedding.
2. Bosons are essential to provide a positive contribution to the supertrace, yielding a minimum mass gap.

3. The supertrace $\mathcal{S}$ serves as the gauge-invariant kernel of the strong force, with its magnitude and entropy determining the invariant mass scale.

4. Off-diagonal matter interactions (radio waves, fermion scattering) perturb the system without violating the invariance, while weak decay emerges as a reduction in the fermionic contribution to $\mathcal{S}$.

Thus, the supertrace unifies the strong nuclear force, mass generation, and thermodynamic irreversibility within a single, geometrically invariant framework.